

A FLEXIBLE RANDOM EFFECTS DISTRIBUTION IN DISEASE MAPPING MODELS

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Summary: Disease mapping has seen many applications in epidemiology and public health. The basic model used in disease mapping is the Besag, York and Mollie model, which incorporates two random effects, one which is spatially structured and the other random effect which is spatially unstructured. The normality assumption on the spatially unstructured random effect is very common. In this work, we investigate a more robust spatially unstructured random effect distribution by considering the symmetric generalized Gaussian distribution in the disease mapping problem. The distribution has the normal and Laplace distributions as special cases. The inferences under this model are carried out under the Bayesian approach implemented in WinBUGS. The generalized Gaussian distribution is introduced in WinBUGS using zero tricks. The usefulness of the proposed model is investigated with a simulation study and applied in real data; mapping tuberculosis in Kenya. In this paper we showed that the generalized Gaussian distribution can produce better results when the normality assumption is violated due to high peakedness or less peakedness in the data. For the case of data in which the random effects are truly normal, the generalized Gaussian distribution adjusts to a normal distribution as dictated by the data itself.

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1. Introduction

Disease mapping refers to the estimation and presentation of summary measures of spatially observed health outcomes. The increased availability of georeferenced data and flexible computational softwares has seen rise in application of disease mapping in the areas of epidemiology and public health, (Rezaeian, Dunn, St Leger and Appleby, 2007; Everitt and Dunn, 2011). Disease mapping can be used to describe geographical variation of diseases, identify clustering of diseases and generate atlas of diseases. A number of statistical reviews on disease mapping have been done (Clayton and Bernardinelli, 1992; Smans and Esteve, 1997; Wakefield, Best and Waller, 2000; Wakefield, 2007; Manda, Feltbower and Gilthorpe, 2011).

The backbone model for univariate disease mapping is the Besag, York and Mollie (BYM) model proposed by Besag, York and Mollie (1991). This model is a form of the generalized linear mixed effects model, with two random effects; a spatially unstructured random effect which is modelled using a normal prior and a spatially structured random effect which is modelled using an intrinsic conditional autoregressive (ICAR) prior. The use of normal distribution to model the spatially unstructured random effects is mainly because of its computational simplicity. Assumption of normality on the uncorrelated random effect in models is common. Sometimes this assumption is incorrect because some random effects can in fact be platykurtic, leptokurtic or skewed; diverging from this general normality assumption, see Box and Tiao (1973). When this normality assumption is violated, there is need to consider other models that would better suit the data at hand. The generalized Gaussian distribution can be used in cases where there is deviation from the normal kurtosis ($kurtosis = 3$) and when there is evidence of skewness in the data. It is a generalization of the common normal distribution to allow for these departures.

The general Gaussian distribution has two versions, both of which add a shape parameter to the normal distribution. The first version of the generalized Gaussian distribution includes normal and Laplace distributions. The continuous uniform distribution arises naturally as a limiting case for this distribution. All the distributions encompassed under this family are symmetric. In the second version, the shape parameter is used to incorporate skewness in the family of distributions. Positive values of the shape parameter produce distributions which are skewed to the left while negative values lead to right skewed distributions. In this work, we concentrate on the symmetric version of the generalized Gaussian distribution. The generalized Gaussian distribution, which we will denote by GGD, has three parameters, the location parameter, μ , the scale parameter, σ^2 and the shape parameter, ϕ . The shape parameter dictates the amount of peakedness or kurtosis.

This work is structured as follows: in section 2, we review the BYM model, in section 3 we introduce the symmetric generalized Gaussian distribution and discuss its limiting distributions, in section 4, we carry out a simulation to study the effect of misspecifying the random effects, in section 5 we use the discussed models to analyze the tuberculosis (TB) data from Kenya and finally discussions and conclusions in section 6.

2. Review of the BYM model

The most commonly used model in single disease spatial analysis was proposed by Besag et al. (1991). It was used to model disease prevalence in regions using the Poisson model. Let λ_i be

the unknown relative risk for region i with respect to a standard population. Also let y_i denote the observed counts of disease in region i and e_i denote the expected count in the same region. The model assumed that the log of relative risk of disease can be broken down into a spatially structured component u_i and a spatially unstructured component v_i . This can be written mathematically as

$$y_i \sim \text{Poisson}(e_i \lambda_i), \quad (1)$$

with

$$\log(\lambda_i) = u_i + v_i, \quad (2)$$

where u_i and v_i are random effects representing unobserved covariates, with u_i representing variables that if were observed would influence the spatial structure, while v_i represents the unobserved heterogeneity in region i . Besag et al. (1991) noted that in most cases, one of the random effects usually dominates the other. If u is stronger than v , then the estimated risk will show spatial structure and if v is stronger than u then the consequence will be to shrink the estimated means towards the overall mean. Besag et al. (1991) assumed that u and v were independent with the following priors:

$$p(v|\tau) \propto \tau^{-\frac{n}{2}} \exp \left\{ -\frac{1}{2\tau} \sum_{i=1}^n v_i^2 \right\}, \quad (3)$$

and

$$p(u|k) \propto k^{-\frac{n}{2}} \exp \left\{ -\frac{1}{2k} \sum_i \sum_{j \in N(i)} (u_i - u_j)^2 \right\}. \quad (4)$$

Basically, equation (3) means that v , the spatially unstructured component is a white noise Gaussian process with unknown variance τ , and equation (4) means that the spatially structured component u , is a Gaussian Markov random field (GMRF) process with variance k , with n being the number of regions under study.

This implies that the conditional distributions of each u_i , given the rest, are given by

$$(u_i|u_{-i}) \sim N \left(\frac{\sum_{j \in N(i)} u_j}{d_i}, \frac{k}{d_i} \right), \quad (5)$$

with

$$E(u_i|u_{-i}) = \frac{\sum_{j \in N(i)} u_j}{d_i} \quad (6)$$

and

$$\text{Var}(u_i|u_{-i}) = \frac{k}{d_i}, \quad (7)$$

where $N(i)$ and d_i are respectively the set and number of neighbours of region i . The neighbourhood can be defined in terms of Euclidean distance of the centroids of the regions or whether two regions share a border or not. This conditional distribution for u is called the intrinsic conditional autoregressive (ICAR) prior distribution. Besag et al. (1991) sampled the posterior distribution using the Gibbs sampler, an MCMC algorithm.

3. The generalized Gaussian Distribution and its properties

Several authors, (Besag et al., 1991; Best, Thomas, Waller, Conlon and Arnold, 1999; Wakefield, 2007), have mentioned that it is possible to replace the normality assumption of the spatially unstructured random effect with either the Laplace distribution or the Student's t distribution. In this work we explore the use of the generalized Gaussian distribution as a candidate for random effects. It has pleasant properties, allowing for the data to dictate the best fitting model for the random effects, whether normal or Laplace adaptively. The generalized Gaussian distribution can result in several interesting distributions upon varying the shape parameter. Of great interest is the fact that the GGD reduces to a normal distribution when the shape parameter has a value of two and to the Laplace distribution when the shape parameter is one.

Definition 1 A random variable X is said to have a GGD if its probability density function is given by

$$f(x; \mu, \sigma, \phi) = \frac{1}{2\Gamma\left(1 + \frac{1}{\phi}\right) \zeta(\phi, \sigma)} \exp\left(-\left|\frac{x - \mu}{\zeta(\phi, \sigma)}\right|^\phi\right) \quad (8)$$

where $x, \mu \in \mathbb{R}$, $\sigma > 0$ and $\zeta(\phi, \sigma) = \left[\frac{\sigma^2 \Gamma(\frac{1}{\phi})}{\Gamma(\frac{3}{\phi})}\right]^{\frac{1}{2}}$. In this expression $\zeta(\phi, \sigma)$ is a scaling factor. See Nadarajah (2005) for further discussions on statistical properties of this distribution.

3.1. Special cases of the GGD

Property 1. If $\phi = 1$ and $\sigma^2 = 2b$, then the pdf of the GGD becomes

$$f(x; \mu, b, 1) = \frac{1}{2b} \exp\left(-\left|\frac{(x - \mu)}{b}\right|\right). \quad (9)$$

Equation (9) is the Laplace probability density function with location parameter μ and scale parameter b .

Property 2. If $\phi = 2$, (8) becomes

$$f(x; \mu, \sigma, 2) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{1}{2} \left(\frac{x - \mu}{\sigma}\right)^2\right). \quad (10)$$

Equation (10) is the probability density function of a normal random variable with mean μ and variance σ^2 .

Property 3. The uniform distribution is a limiting case of the GGD when $\phi \rightarrow \infty$.

4. Simulation

In this section, we carry out a simulation study to determine the effect of wrongly specifying the distribution of the random effect in a BYM model. Three scenarios were considered in the simulation.

In the first simulation, the datasets is generated through a random effect, v with a peaked kurtosis as follows: Assuming that there are 60 geographical regions and O_i is the number of disease counts observed in region i and E_i is the corresponding expected counts in that region. Without loss of generality, we further assume that no covariates are available for use.

1. Step 1: Generate 60 values of $v \sim GGD(0, 0.1, 1.02)$, which is platykurtic.
2. Step 2: Set $E = 40$ for all the regions.
3. Step 3: Calculate the relative risk, $\theta = \log(v)$
4. Step 4: Calculate $\lambda = E \times \theta$
5. Step 5: Generate the observed counts as $O \sim \text{Poisson}(60, \text{mean} = \lambda)$.

We fitted two Bayesian hierarchical models for the data set. The models were specified based on different assumptions on the random effects as follows:

$$O_i \sim \text{Poisson}(\mu_i) \quad (11)$$

and

$$\log(\mu_i) = \log(E_i) + v_i \quad (12)$$

with

- Model a: $v_i \sim GGD(0, \sigma_1^2, \phi)$,
- Model b: $v_i \sim N(0, \sigma_2^2)$.

The estimated relative risk $\hat{\theta}_i = \log(v_i)$.

The simulation steps above were repeated $m = 1000$ times.

To compare the two models, we calculated the mean squared error (MSE), for each model, using the formula:

$$MSE = \frac{1}{1000} \sum_{j=1}^{1000} \frac{1}{60} \sum_{i=1}^{60} (\hat{\theta}_{ij} - \theta_{ij})^2. \quad (13)$$

The model with a small MSE provides the best fit.

In the second scenario, the procedure above is repeated but changing the random effect generation mechanism in step 1, as $v \sim N(0, 0.1)$.

Similarly, in the third scenario, the procedure was repeated with the random effect being generated using $v \sim GGD(0, 0.1, 12)$, yielding platykurtic random effect.

From the simulation results, the generalized Gaussian random effect is seen to adapt well even in cases where the random effect strictly follows a normal distribution. In Table 1, the generalized Gaussian distribution produces lower mean squared error values as compared to the normal distribution in all the cases. When the random effects are platykurtic, the loss in efficiency incurred for using normal random effects is 17.841% and when the random effects are leptokurtic, the percentage loss in efficiency is 17.822%. When the random effects are generated using normal distribution, still the GGD has a lower mean squared error compared to the normal counterpart. The loss in efficiency for this case is low at 6.19%.

Table 1: Model comparison under simulation

Generating dist.	parameters used	model used	MSE	% loss in efficiency
GGD(Leptokurtic)	$\mu = 0, \sigma^2 = 0.1, \phi = 1.02$	Normal	0.006497	17.8
		GGD	0.005514	0
Normal	$\mu = 0, \sigma^2 = 0.1$	Normal	0.003297	6.2
		GGD	0.003093	0
GGD(Platykurtic)	$\mu = 0, \sigma^2 = 0.1, \phi = 12$	Normal	0.002926	17.8
		GGD	0.002483	0

5. Application: Mapping of tuberculosis in Kenya

In this section we apply the model to TB data collected by the Ministry of Health, Kenya. The division of leprosy, TB and lung disease (DLTLD) is responsible for the data collection within this ministry. This central unit receives case finding reports from all counties on a quarterly basis and aggregates the values for the whole year. Table 3 in Appendix A summarises the number of TB cases per county and the corresponding population estimates for the year 2002.

The following models, with increasing complexity, were fitted:

$$O_i \sim \text{Poisson}(\mu_i) \quad (14)$$

with

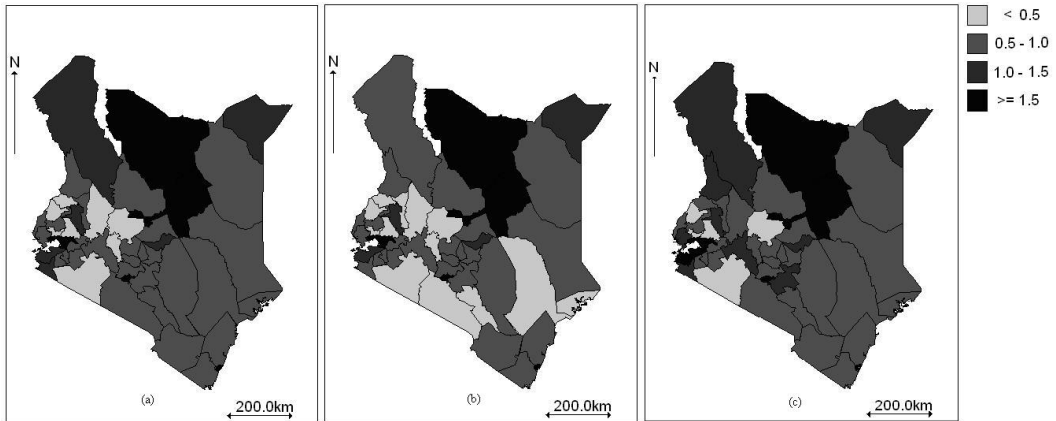
- Model 1: $\log(\mu_i) = \log(E_i) + \beta_0 + v_i; \quad v \sim N(0, \sigma_v^2)$
- Model 2: $\log(\mu_i) = \log(E_i) + \beta_0 + v_i; \quad v \sim \text{GGD}(0, \sigma_v^2, \phi)$
- Model 3: $\log(\mu_i) = \log(E_i) + \beta_0 + u_i; \quad u \sim \text{ICAR}$
- Model 4: $\log(\mu_i) = \log(E_i) + \beta_0 + v_i + u_i; \quad v \sim N(0, \sigma_v^2), u \sim \text{ICAR}$
- Model 5: $\log(\mu_i) = \log(E_i) + \beta_0 + v_i + u_i; \quad v \sim \text{GGD}(0, \sigma_v^2, \phi), u \sim \text{ICAR}$

where O is the observed counts of cases of TB and E is the expected count of cases of TB. Model estimation was carried out using a Bayesian approach. All parameters in the models were assigned prior distributions. In this analysis, a non-informative normal prior was assigned to the fixed effect coefficient β_0 , the shape parameter ϕ was given a diffuse, uniform prior, and the variance parameters were assigned inverse gamma distributions. The models were implemented using WinBUGS version 1.4 (Spiegelhalter, Thomas, Best and Lunn, 2007; Ntzoufras, 2011). For each model, 50,000 Markov chain Monte Carlo (MCMC) iterations were ran, with the initial 10,000 discarded to cater for the burn-in period and thereafter keeping every tenth sample value. The 4,000 iterations left were used for assessing convergence of the MCMC and parameter estimation. We assessed MCMC convergence of all models parameters by checking trace plots and autocorrelation plots of the MCMC output, see Gelman, Carlin, Stern and Rubin (2003). The models were compared using the Deviance Information Criterion (DIC) as suggested by Spiegelhalter, Best, Carlin and Van Der Linde (2002).

Table 2: Model comparison in mapping TB in Kenya

Model	β_0	ϕ	Estimates		pD	DIC
			σ_u^2	σ_v^2		
Model 1	-0.21(-0.31,-0.08)	-	-	0.07(0.03,0.19)	46.48	510.64
Model 2	-0.20(-0.38,-0.02)	2.27(1.07,6.27)	-	0.25(0.18,0.47)	45.76	507.81
Model 3	-0.22(-0.22,-0.19)	-	1.31(0.59,2.69)	-	49.76	515.12
Model 4	-0.22(-0.39,-0.11)	-	0.02(0.01,0.13)	0.06(0.03,0.11)	46.98	511.35
Model 5	-0.18(-0.30,-0.02)	3.92(0.82,7.83)	0.31(0.00,1.56)	0.16(0.00,0.34)	50.55	519.47

The best fitting model is one with the smallest DIC value. In this analysis, the unstructured heterogeneity, modelled using the generalized Gaussian distribution was found to perform slightly better than the other models considered in this study. This can be seen in Table 2, based on the DIC values. Figure 1 shows the spatial distribution of TB in Kenya based on this best fitting model. This is a map of relative risk and its corresponding credible interval.

Figure 1: TB relative risk map(a) and the corresponding 95% lower(b) and upper(c) credible limits maps, respectively, produced by model 2.

6. Discussion

Routine framework for modelling correlated data is through the generalized linear mixed effects model in which a random effect is incorporated. The usual main assumption in a standard version of the setup is to model the between subject variations with random effects that are normally distributed. The assumption of modelling random effects with a normal distribution has been both challenged and supported by several authors (McCulloch and Neuhaus, 2011; Litière, Alonso and Molenberghs, 2007; Litière, Alonso and Molenberghs, 2008). A lot of work has been done in trying to find better fitting distributions in the recent past. Magder and Zeger (1996) proposed a smooth non-parametric maximum likelihood approach to modelling the random effects. Verbeke and Lesaf-

fre (1997) proposed using a mixture of normal distributions for the random effects and they carried out their estimation using the expectation maximization (EM) algorithm. Zhang and Davidian (2001) proposed a semi-parametric linear mixed model in which they assumed that the random effects have a smooth density represented by semi-nonparametric truncated series expansion. Ho and Hu (2008) used a finite mixture of normal in a Bayesian setting with the number of components being estimated from the data automatically.

In disease mapping context, the same situation arises. The basic BYM model has two components, one which is spatially structured and the other component which is spatially unstructured. The spatially unstructured component is usually modelled using the normal distribution. In this work we propose the generalized Gaussian distribution as a random effect distribution to replace the over-restrictive normal distribution for the unstructured heterogeneity. The special cases of the generalized Gaussian distribution, including the normal and Laplace distributions, are exposed. The generalized Gaussian distribution has an extra parameter to allow for high and low peakedness as dictated by the data.

The parameters in the models are estimated under Bayesian inference. The models were implemented in WinBUGS. The generalized Gaussian distribution is not a standard distribution in the WinBUGS software. We introduced this distribution in the software using zero tricks, see Appendix B. The models were compared using simulation studies and again with a real data set.

In the simulation study it was seen that the effect of misspecification of the random effects when the normal distribution is used in place of the generalized Gaussian distribution was high as compared to using the generalized Gaussian in place of the normal distribution. The generalized Gaussian distribution has all the nice properties of the normal distribution. In fact the normal distribution is a special case of the generalized Gaussian distribution. When the random effect distribution fails to adhere to the normality assumption due to peakedness, the generalized Gaussian distribution plays a big role in capturing this, something that the normal distribution cannot.

In the real data sets comparison, the generalized Gaussian distribution is seen to perform better than the normal distribution model. This model was used to produce county specific maps of relative risk of TB in Kenya. The maps are critical in understanding disease epidemiology and also in helping policy makers to develop informed intervention programs and allocate scarce resources adequately.

One limitation of this model is that it only captures high and low peakedness departures from the normal distribution. It assumes that the random effects are symmetric. This assumption can at times also be wrong. More flexible random effects models, which can also capture skewness can be investigated.

Appendices

A Tuberculosis data

Table 3: Number of TB cases reported for each county and the corresponding population size.

County	TB cases	Population	County	TB cases	Population
Baringo	572	461175	Mandera	993	286006
Bomet	729	437321	Marsabit	989	194960
Bungoma	1343	1134381	Meru	2380	1221068
Busia	1262	618068	Migori	1974	746904
Elgeyo Marakwet	395	326798	Mombasa	5889	755867
Embu	1145	497662	Muranga	1541	808488
Garissa	1000	480489	Nairobi	15979	2495170
Homa Bay	3159	829355	Nakuru	3413	1354899
Isiolo	611	107741	Nandi	720	659957
Kajiado	613	461174	Narok	610	604298
Kakamega	1979	1454722	Nyamira	763	548053
Kericho	1936	890544	Nyandarua	639	526742
Kiambu	2638	1523061	Nyeri	1536	722739
Kilifi	1521	923837	Samburu	340	163001
Kirinyaga	721	502243	Siaya	2034	790555
Kisii	2105	1052456	Taita Taveta	569	279951
Kisumu	4753	882705	Tana River	278	195965
Kitui	2166	908106	Tharaka-Nithi	1053	338616
Kwale	945	559901	Trans Nzoia	876	652005
Laikipia	344	365759	Turkana	1340	516833
Lamu	111	83985	Uasin Gishu	2384	707664
Machakos	2474	1005586	Vihiga	515	561538
Makueni	1119	856800	Wajir	743	377527
			West Pokot	915	349857

B Specifying the GGD prior in WinBUGS using zero tricks

Since the GGD prior is not specified as a standard distribution in WinBUGS, we used the zero trick technique to specify this prior distribution. Suppose we wish to use a prior $GGD(\theta)$ which is not available, we first define a flat/non-informative prior for θ , say, $h(\theta)$. Next we define a dummy variable, say *Zeros*, with all its values set to zero. We also specify this dummy variable, *Zeros*, to be following a Poisson distribution with mean λ . Then we set λ to be equal to the negative log-

likelihood of the prior distribution that we are interested in, the GGD, that is $\lambda = -\log(GGD(\theta))$.

$$\begin{aligned}
 f(\theta|Zeros) &= f(Zeros = 0|\theta)h(\theta) \\
 &= \frac{e^{-\lambda} \lambda^{Zeros}}{Zeros!} h(\theta) \\
 &= \frac{e^{-\lambda} \lambda^0}{0!} \times 1 \\
 &= e^{-\lambda} \\
 &= e^{\log(GGD(\theta))} \\
 &= GGD(\theta)
 \end{aligned}$$

This is the prior distribution that that we wanted. The corresponding code in Winbugs software is given below.

```
#Generalized Gaussian prior distribution implementation using zero tricks
my_zeta<-pow(abs((sigmau*sigmau*exp(loggam(1/psi)))/(exp(loggam(3/psi))))),0.5)
normalizing<-1/((2*exp(loggam(1+1/psi)))*my_zeta)
logN<-log(normalizing)
for(i in 1: n)
{
v[i]~dunif(-10000,10000)
zeros[i]<-0
logGGD[i]<-logN-pow(abs(v[i]/my_zeta),psi)
zeros[i]~dpois(-logGGD[i])
}
```

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